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DEPARTMENT: CSE AIML **TASK:** 30-Minute Student Practical

BUILD YOUR OWN RULE-BASED PYTHON CHATBOT

AIM

To create a simple rule-based Python chatbot that interacts continuously with the user, understands different messages, provides suitable responses, and exits when requested.

PYTHON PROGRAM

```
print("Hello! I am your Study Buddy Chatbot.")

name = input("What is your name? ")

print("Nice to meet you, " + name + "!")
print("I can talk about college, programming, and studies.")
print("Type 'bye' or 'exit' whenever you want to stop.")

while True:
    message = input(name + ": ").lower().strip()

    if "bye" in message or "exit" in message:
        print("Bye " + name + "! Keep learning and have a great day!")
        break

    elif "hello" in message or "hi" in message:
        print("Hi " + name + "! What would you like to know?")

    elif "college" in message:
        print("College is a great place to learn new skills and make good
friends.")

    elif "programming" in message or "coding" in message:
        print("Programming means writing instructions for a computer to perform a
task.")

    elif "study" in message or "exam" in message:
        print("A good study plan, regular practice, and revision can help you
prepare well.")

    elif "your name" in message:
        print("I am Study Buddy, your simple Python chatbot.")

    else:
        print("I am still learning. Please ask me about college, programming, or
studies.")
```

EXPLANATION

1. The program first asks the user to enter their name.

2. The chatbot uses the user's name while responding.
3. `while True` keeps the conversation running continuously.
4. `input()` receives messages from the user.
5. `lower().strip()` cleans the user's input.
6. The chatbot recognizes greetings such as hello and hi.
7. It can respond to questions about college, programming, and studies.
8. `if / elif / else` is used for decision making.
9. The `in` operator allows variations such as "hello bot" or "I like coding".
10. `break` ends the conversation when the user types bye or exit.
11. The `else` statement gives a useful response when the chatbot does not recognize the message.

SAMPLE CONVERSATION

```
Hello! I am your Study Buddy Chatbot.
What is your name? Priya

Nice to meet you, Priya!
I can talk about college, programming, and studies.
Type 'bye' or 'exit' whenever you want to stop.

Priya: hello bot
Hi Priya! What would you like to know?

Priya: what is programming
Programming means writing instructions for a computer to perform a task.

Priya: tell me about studies
A good study plan, regular practice, and revision can help you prepare well.

Priya: what is college
College is a great place to learn new skills and make good friends.

Priya: xyz
I am still learning. Please ask me about college, programming, or studies.

Priya: bye
Bye Priya! Keep learning and have a great day!
```

TEST CASES

Input	Expected Response
hello	Greeting response
HELLO	Greeting response
hi	Greeting response
hello bot	Greeting response

Input	Expected Response
what is programming	Programming response
tell me about college	College response
how should I study	Study response
what is your name	Chatbot name response
unknown message	Useful unknown-message response
bye	Exit response
exit	Exit response

CONCLUSION

The Study Buddy chatbot successfully follows the required conversational flow: INPUT → PROCESS → RESPOND → REPEAT → EXIT. It continuously interacts with the user, recognizes different topics, responds using predefined rules, and terminates when the user enters bye or exit.